COMET
Generative Medical Event Models Improve with Scale

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BACKGROUND

Real-World Data (RWD)

- Data generated during routine clinical care, outside controlled clinical trial settings.

  Major Sources:
  - Electronic Health Records (EHR/EMR): hospital clinical data
  - Claims Data: insurance billing and reimbursement records
  - Patient-Generated Data: wearable devices, activity tracking, symptom logs
  - Disease Registries: condition-specific longitudinal patient databases (e.g., cancer, rare diseases)

Real-World Data (RWE)

- Clinical evidence derived from the statistical analysis of RWD.

  Key Applications:
  - Post-Marketing Safety Surveillance (PMS): detecting adverse events after drug approval
  - Label Expansion: evaluating effectiveness in new indications
  - Regulatory & Reimbursement Decisions: supporting policy and coverage determinations

BACKGROUND

Limitations of the Current RWD → RWE Pipeline

Still depends on:

- Manually selecting eligible patients
- Developing separate models for each question
- Heavy reliance on expert-driven analysis

Emergence of the Epic Cosmos Platform

- 300M+ patients, 16B+ encounters
- Unified longitudinal records across health systems
- Rich clinical data: diagnoses, lab results, medications, genomics and social determinants of health

Introducing CoMET

- **A foundation model trained on Cosmos to enable zero-shot clinical prediction.**

CoMET: Methods

(1) Data Pre-processing Pipeline

16.3B Encounters / 300M+ Patients → Patient & encounter selection + post-filter clean up → 118M Unique Patients

90% for Training
10% for Evaluation

(2) From Patient Records to Discrete Tokens

Event Type	Token (Example)
Demographics	[SEX_MALE] [RACE_WHITE] [AGE_62-66]
Encounter Start	[OFFICE_VISIT_START] [DEPT_FAMILY_MED]
Chief Complaint	[CC_FATIGUE]
Lab Result (HbA1c 8.6%)	[LOINC_HbA1c] [Q_8-9]
Diagnosis (Type 2 Diabetes)	[DX_E11] [DX_9]
Medication (Semaglutide)	[ATC_A10] [ATC_BJ] [ATC_06]
Time Passage	[TIME_2-3MONTHS]

CoMET: Methods

(3) Model architecture setting

- Decoder-only transformer (based on Qwen2)
- Random initialization
- Three compute-optimal models: S (small) / M (medium) / L (large) (determined by a scaling-law analysis)

Name	Parameters	Training tokens	Layers	Dimension	MLP† Dimension	Compute (TFLOPs†)
CoMET-S	62M	90B	6	768	3072	67M
CoMET-M	119M	160B	12	768	3072	234M
CoMET-L	1B	1.7T	16	2048	8192	14B

How was it trained?
- Cross-entropy next-token prediction
- AdamW optimizer with cosine learning rate decay
- Sequences densely packed (batch size 512)

What do we compare it to?
- Logistic / Linear regression (BoW features)
- XGBoost (strong structured-data baseline)
- Supervised transformer (same architecture, no pretraining)

† MLP: Multi-layer perceptron

† TFLOP: tera floating point operation

CoMET: Methods

(4) Inference

1. Generate Future Trajectories

Patient history observed up to time t_0 $\xrightarrow[\text{generated}]{\text{Autoregressively}}$ $y_{i1}, y_{i2}, \dots, y_{id}$

- y_{ij}: j-th generated medical event
- Repeat n times (Monte-Carlo sampling)

2. Assign Time to Generated Events

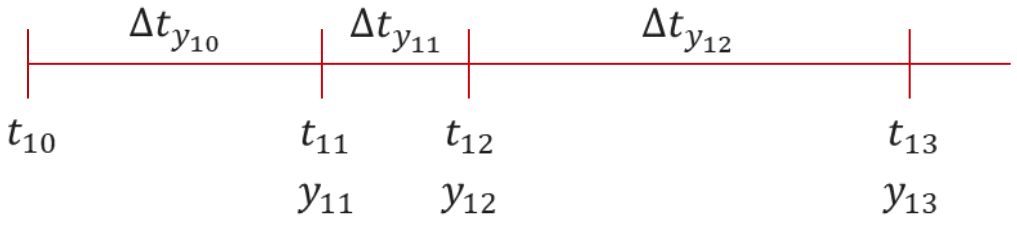
- Each token has a time increment Δt_y

$$t_{ij} = t_{i(j-1)} + \Delta t_{y_{i(j-1)}} \quad i = 1, \dots, n$$

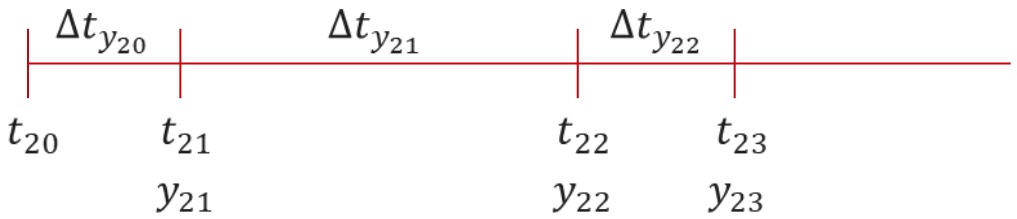
So each simulation becomes:

$$T_i = \{(y_{i1}, t_{i1}), \dots, (y_{id}, t_{id})\}$$

Simulation 1



Simulation 2



where Δt_y = geometric midpoint of the token's time interval

CoMET: Methods

(4) Inference

3. Estimate Risk Within Time Horizon τ

Let S be a target event set (e.g., hospitalization)

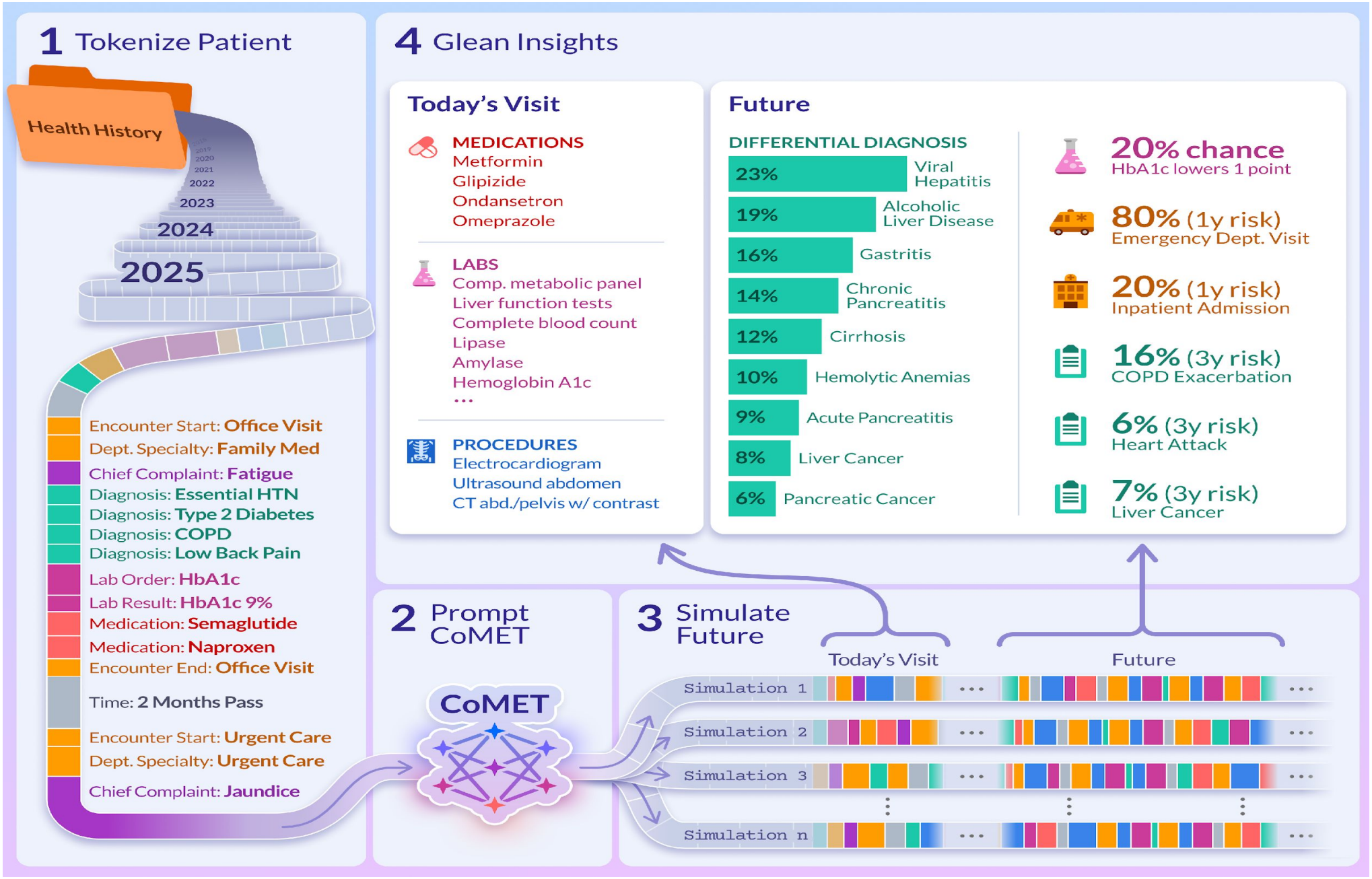
$$P_\tau(S) = \frac{\#\text{ simulations where } S \text{ occurs before } \tau}{\#\text{ simulations that reach } \tau}$$

Right-censoring applied

Example. $\tau = 1$ year

	Hospitalization time	Counted
1	8 month	1
2	No event	0
3	14 month	
4	Stopped at 6month	

CoMET: Overview



CoMET: Evaluation

- Is it realistic?
- Can it reproduce clinical encounters?
- Does it predict disease risk accurately?
- …

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Metrics

- **AUCROC:** Discrimination ability of risk prediction

$$\mathrm{AUCROC} = \Pr(\hat{r}_{\mathrm{diseased}} > \hat{r}_{\mathrm{healthy}})$$

- $\hat{r}_{\mathrm{diseased}}$: predicted risk for a truly diseased patient
- $\hat{r}_{\mathrm{healthy}}$: predicted risk for a non-diseased patient
- 0.5: random / 1.0: perfect separation

- **MAE:** Forecasting accuracy

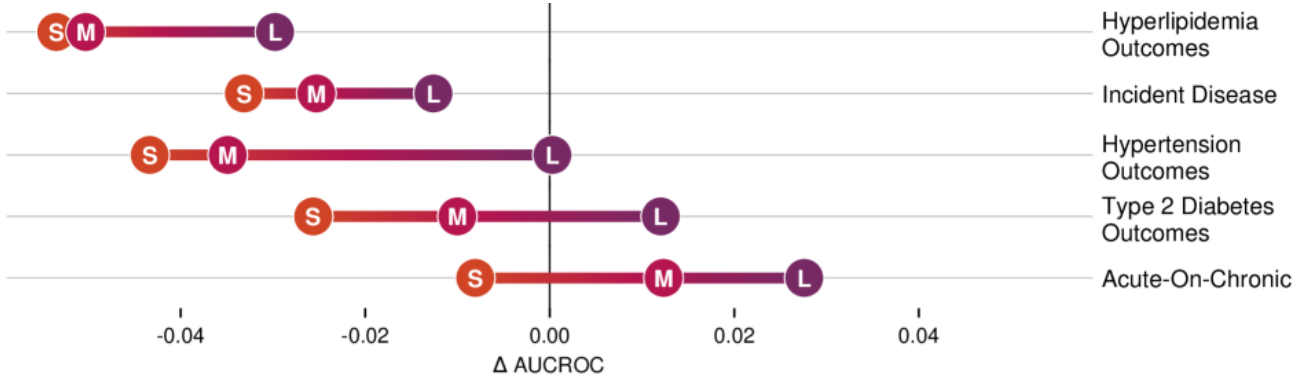
$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N} | y_i - \hat{y}_i |$$

- y_i: true event count
- $\hat{y}_i$: predicted event count
- N: Number of patients

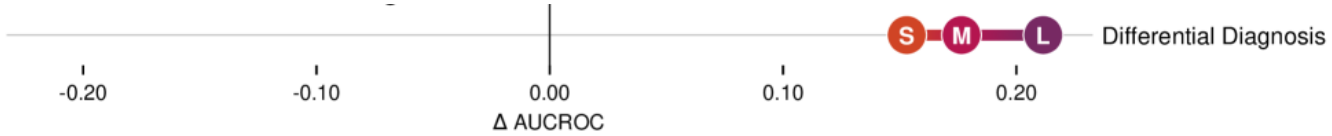
➢ **PR-AUC** (precision-recall area under the curve), **RMSLE** (root-mean squared log error) , …

CoMET: Selected Result

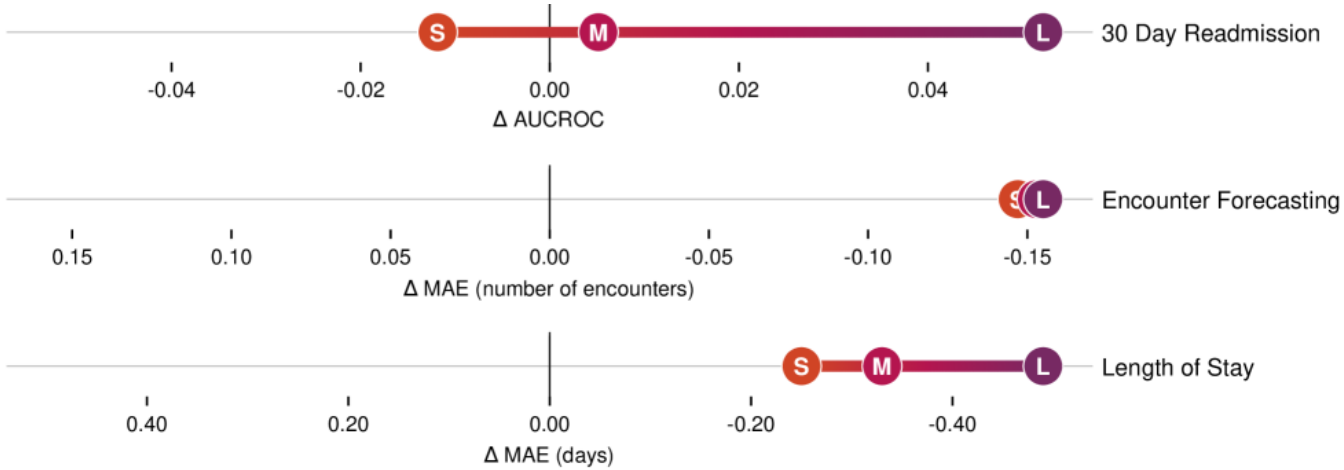
Disease Risk



Differential Diagnosis

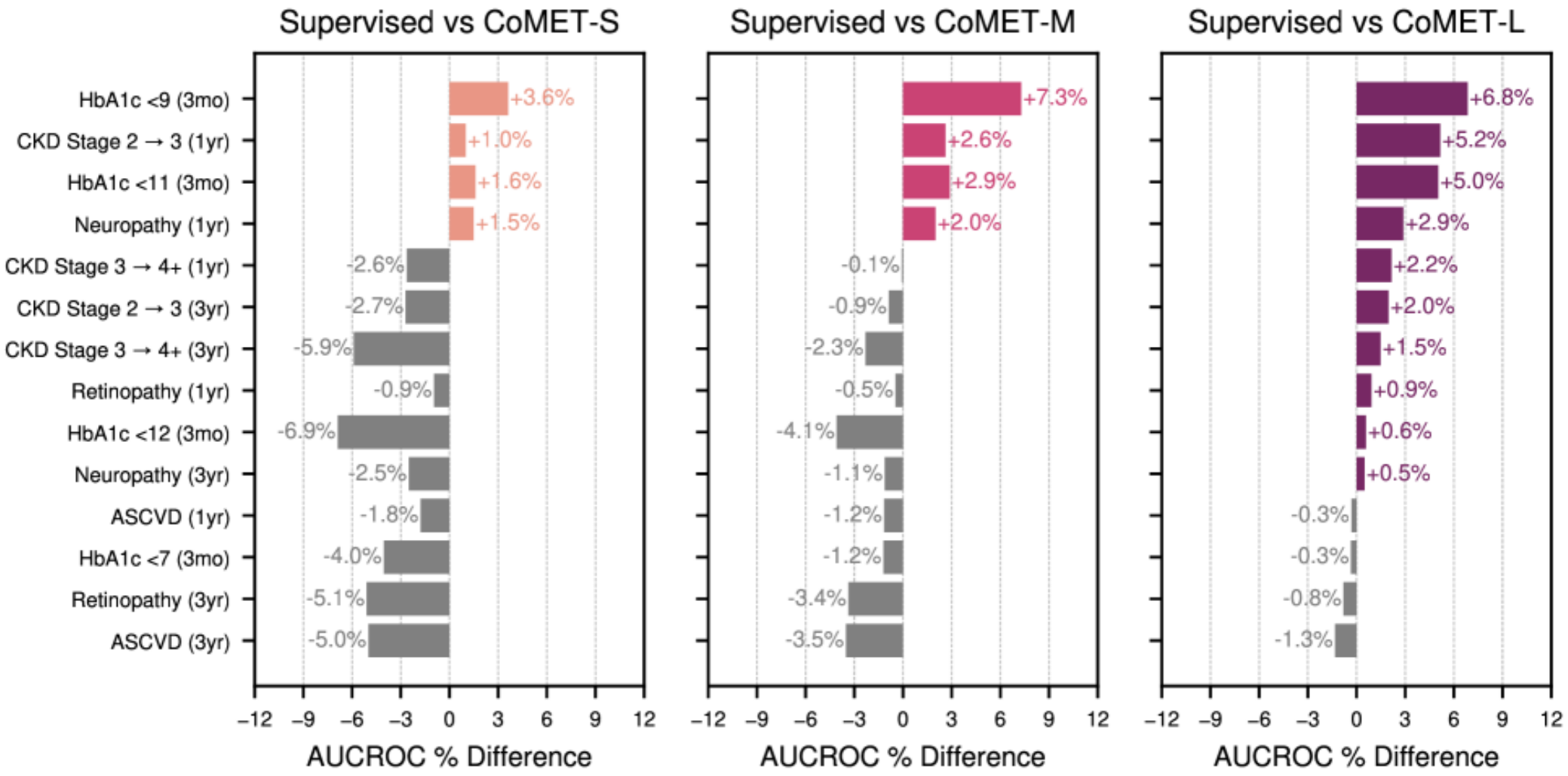


Patients' interactions With the Health System



CoMET: Selected Result

Outcome prediction at medication initiation (T2DM patients)



% Difference = (CoMET's AUCROC – Baseline Model's AUCROC) / Baseline Model's AUCROC

Summary

Large-Scale Foundation Model

- Trained on Epic Cosmos (118M+ patients)
- Decoder-only transformer (up to 1B parameters)

Learns Patient Journeys

- Processes medical events as a time-ordered sequence
- Uses diagnoses, medications, labs, procedures, and time information

Generative Prediction

- Simulates possible future health trajectories
- Estimates future risks probabilistically

Zero-Shot Capability

- No task-specific fine-tuning required
- One model supports many clinical tasks

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